EXSU 620 / PROJECT 02 / DESIGN DOSSIER

Continuous endotracheal cuff pressure monitor

A bench instrument that measures endotracheal cuff pressure between manometer checks, separates a cough or suction pass from a leak, and alerts the care team when the cuff has stayed outside the safe window long enough to matter.

GROUP	Group 2, Project 02, cuff pressure, continuously
AUTHORS	Karman Kang, Nathalie Obi, Imen Benadda, Kayla Arab, Lucy Han
DEVICE	Cuff pressure monitor, bench prototype, revision B (firmware 2026-11-05 v8, alert service v1)
MEASURAND	Cuff pressure in cmH2O, from absolute pressure in a sealed air chamber referenced to room air at start-up
REFERENCE	Dial manometer taken from a manual arm blood pressure cuff set, connected through the second three way stopcock
STATUS	Live pressure measurement working on the board. Alarm rules and the Wi-Fi dashboard tested in two bench runs on 3 November 2026, with 68 and 91 samples exported as CSV. Dashboard revision B, the cough and suction labels, the SQLite store and the text message path tested in a desktop simulation only. Physical ring, speaker and button not built; the dashboard replaces them. All gaps are declared in sections 9 to 11.

> **How to read this document.** Every hardware result comes from what was built and observed between 22 October and 3 November 2026. Results from the desktop simulation are labelled as simulation. Nothing has been entered as measured that was not measured.
> **Supporting files.** This PDF is submitted inside EXSU620_Group2_Submission.zip. References in the form `[zip: folder/file]` point to the file in that zip; Appendix A lists every file.

SECTION 1

Clinical requirements, as issued

ID	REQUIREMENT
CR-1	Cuff pressure shall be known continuously and reported in cmH2O, without deflating the cuff, disconnecting the circuit, or waking the patient.
CR-2	The care team shall be alerted when cuff pressure stays outside 20 to 30 cmH2O for more than 30 seconds. An event shorter than 5 seconds shall not raise an alarm.
CR-3	A falling pressure consistent with a cuff or pilot line leak shall be reported as a leak, separately from a low pressure reading, because the clinical response differs.
CR-4	Time spent inside 20 to 30 cmH2O over a shift shall be recorded, together with every excursion, its duration, and who acknowledged it.

SECTION 2, DELIVERABLE D1

Engineering requirements

Each engineering requirement states a range, rate, latency or accuracy, names the clinical requirement it serves, and carries an acceptance criterion that a test can pass or fail.

ID	TRACES TO	REQUIREMENT	ACCEPTANCE CRITERION
ER-1	CR-1	Report cuff pressure in cmH2O at 1 Hz or faster over 0 to 60 cmH2O, with a display resolution of 0.1 cmH2O.	Pass if a 90 s log shows a sample rate within 10 percent of nominal and reported values in steps of 0.1 cmH2O or less.
ER-2	CR-1	Agree with the reference manometer within 2 cmH2O over 10 to 50 cmH2O.	Pass if the difference between sensor and manometer readings does not exceed 2 cmH2O at each recorded comparison point.
ER-3	CR-1	Maintain a stable room air baseline, with variation under 1 cmH2O during a short recording after zeroing.	Pass if baseline variation stays under 1 cmH2O over at least 15 s with the line open to room air.
ER-4	CR-2	Raise an advisory when pressure stays outside 20 to 30 cmH2O for 30 s, and a critical alarm when it then stays below 15 or above 35 cmH2O for 30 s.	Pass if advisory and critical transitions occur within 2 s of their 30 s dwell conditions.
ER-5	CR-2	An excursion shorter than 5 s, upward (cough) or downward (suction), shall raise no alarm.	Pass if short pressure fluctuations under 5 s do not change the alarm state.
ER-6	CR-3	Report a falling pressure faster than 2 cmH2O per minute as a leak, separately from a low pressure alarm.	Pass if a staged leak is labelled as a leak and not as low pressure.

ER-7	CR-3	Detect a technical condition (sensor not answering, flat signal, line open to room air, zero not set) within 10 s and never show a normal state while it persists.	Pass if the technical alarm activates within 10 s of the condition, with no false normal indication.
ER-8	CR-4	Record pressure, alarm events, acknowledgements and the acknowledging user with timestamps, and export them as CSV.	Pass if the CSV export contains pressure readings, alarm events and acknowledgement times matching the event records within 2 s.

ER-3 is written because the BMP280 measures absolute pressure. The device subtracts a room air baseline taken at start-up, so any later change in room pressure appears as a change in cuff pressure. A weather change of 5 hPa over a shift is about 5 cmH2O, half the width of the safe window. Only short term stability was assessed (section 9).

SECTION 3, DELIVERABLE D2

System design

Final bench configuration. The pneumatic side is built from two three way stopcocks connected in series. The first stopcock (A) connects the sealed air chamber that holds the BMP280 sensor, a 200 cc syringe used for air entry and exit and for balloon inflation, and the second stopcock. The second stopcock (B) connects, through tubing, to the balloon inside the PVC trachea phantom, and through a second tubing to a dial manometer taken from a manual arm blood pressure cuff set. The levers select which branches share air, so the same closed volume can be read by the sensor and by the manometer, or the manometer can be turned out of the circuit.

> **Resources received.** The project card lists a cuffed endotracheal tube, syringe, three way stopcock, pilot line tubing and a clinical cuff manometer as part of every group's kit. This group received only a balloon, a trachea phantom and a manual arm blood pressure cuff with a dial manometer. No tubing, syringes or stopcocks were provided; the team bought them out of pocket in order to build a prototype at all. The dial manometer has no electronic output, so its readings cannot be logged against the sensor without building a significant extra system, such as a camera that films the dial and reads the needle. These limits come from the equipment provided, were outside the team's control, and are not failures of the design or rapid prototyping process. They reflect unequal access to the equipment listed on the card and should be weighed as such when requirements that depend on them are assessed (ER-2, CR-1).

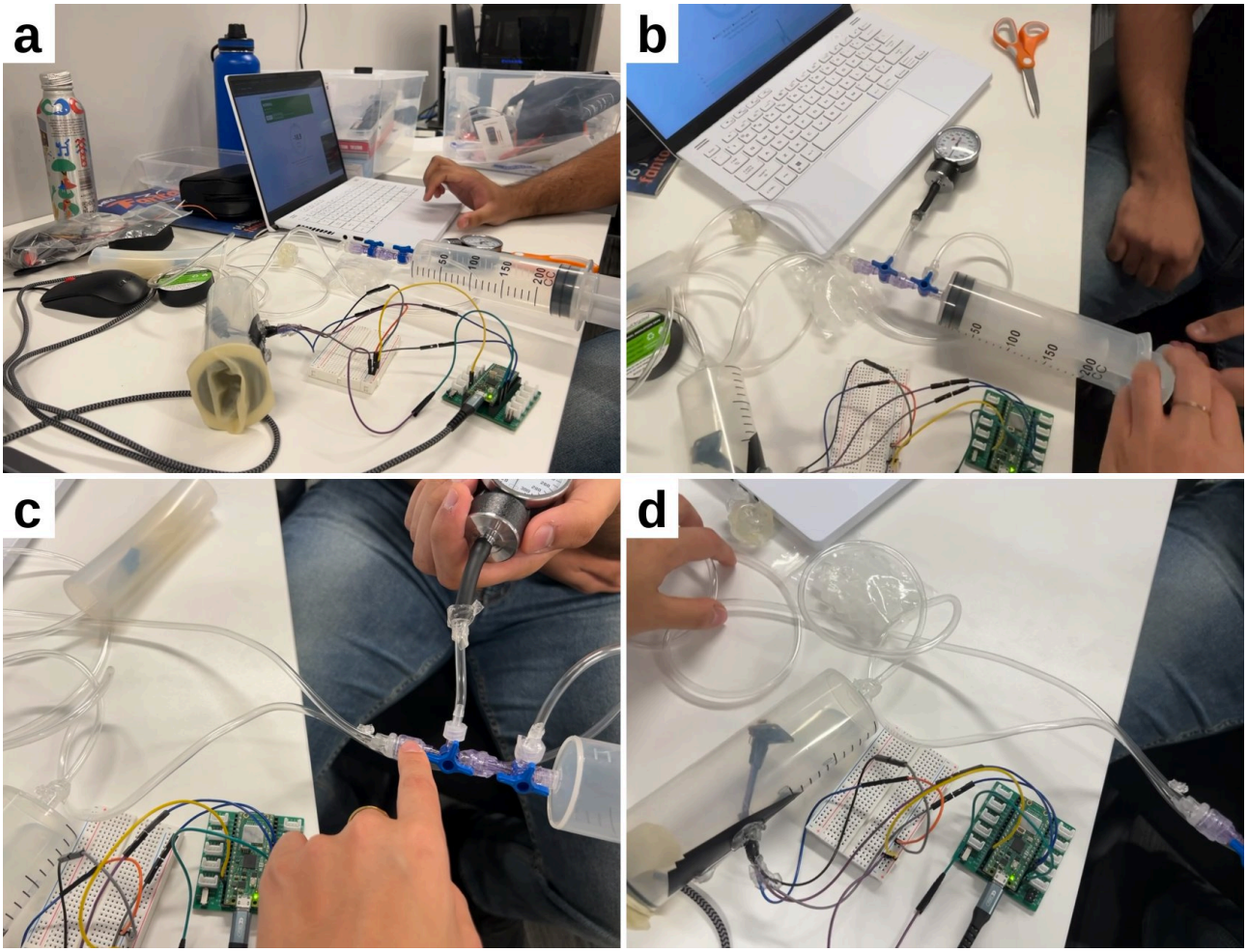


Figure 1. Final bench configuration, 3 November 2026. (a) Whole set-up: PVC trachea with the balloon, sealed sensor chamber, breadboard and Pico 2 W on the Grove shield, the two stopcocks with the 200 cc syringe, and the laptop showing the dashboard. (b) Stopcock A (syringe side) and stopcock B in series, with the dial manometer on the branch tube of stopcock B. (c) Close-up of the two blue three way stopcocks: manometer line above, 200 cc syringe on the right, tubing to the balloon and to the sensor chamber on the left. (d) Sealed chamber with the BMP280 inside and its I2C wires to the breadboard, with the tubing to the trachea balloon. Full resolution image: `[zip: docs/final_rig.jpg]`

Why this design. The first build followed Figure 1 of the project card, with a single stopcock teed into the pilot line and a 10 mL syringe. It had air leaks in the tubing that the team could not resolve. With two stopcocks and a 200 cc syringe, the team could draw air back in and re-inflate the balloon whenever pressure was lost, and keep a live pressure reading on the dashboard throughout. The second stopcock also gave a free port for the dial manometer, so the reference could be read on the same volume as the sensor.

Electronics and software. The Raspberry Pi Pico 2 W reads the sensor over I2C on GP4 (SDA) and GP5 (SCL), takes the room air baseline during the first 2 s after start-up, and converts the difference to cmH2O (1 cmH2O = 98.07 Pa). The alarm rules of Figure 3 run on the Pico. The Pico creates its own Wi-Fi network (CuffMonitor) and serves the dashboard at 192.168.4.1, so any laptop or tablet that joins the network sees the nurse view without internet or a school network. The dashboard gives the visual and audible alarms and the acknowledge control, replacing the planned

NeoPixel ring, speaker and button. During development the same firmware streams values to the Thonny plotter over USB. Pressure and alarm events are exported as CSV from the dashboard.

OPTION CONSIDERED	DECISION	REASON
Full Raspberry Pi, as in the building plan	Not applicable	The kit board is a Pico 2 W. The building plan's pin table and operating system set-up apply only to a full Raspberry Pi.
Single stopcock teed into the pilot line, 10 mL syringe, as in card Figure 1	Rejected after testing	Air leaks in the tubing could not be sealed, so pressure could not be held long enough to read.
Two three way stopcocks in series with a 200 cc syringe	Selected	Air can be drawn back in and the balloon re-inflated during a run, and the second stopcock gives a port for the manometer. The 200 cc syringe is not on the resource list; this row is its design rationale.
Clinical cuff manometer as reference	Substituted	A dial manometer from a manual arm blood pressure cuff set was used. It reads mmHg (1 mmHg = 1.36 cmH2O) and starts at 20 mmHg (27 cmH2O), which limits the comparison below the safe window (section 11).
Sensor on the shield's SPI0 header, all 7 wires	Rejected	More wiring and connection complexity. Replaced by 4 wires on the I2C bus.
Sensor on a Grove port with an adapter cable	Not used	The Grove to jumper cable was not available at the time of wiring.
Sensor on the I2C bus, GP4 and GP5, 4 wires	Selected	Standard wiring for this sensor, fewest wires.
NeoPixel ring, speaker and button at the bedside	Rejected after testing	The ring stayed dark after several rewiring attempts (FM-1, FM-2). Alarms moved to the dashboard.
Laptop service with SQLite, fed over USB, as in the card	Deferred	The SQLite service and phone notifications were not built. The Pico's dashboard and CSV export were used instead.
Pico joining the school Wi-Fi	Rejected	Device registration and network rules are outside the team's control. The Pico's own network works anywhere.
High pressure alarm that sounds instantly above 30 cmH2O	Rejected	Fails CR-2, which forbids alarms on events under 5 s such as a cough. A 30 s dwell is used before the advisory.

SECTION 4, DELIVERABLE D3

Component, module and system chart

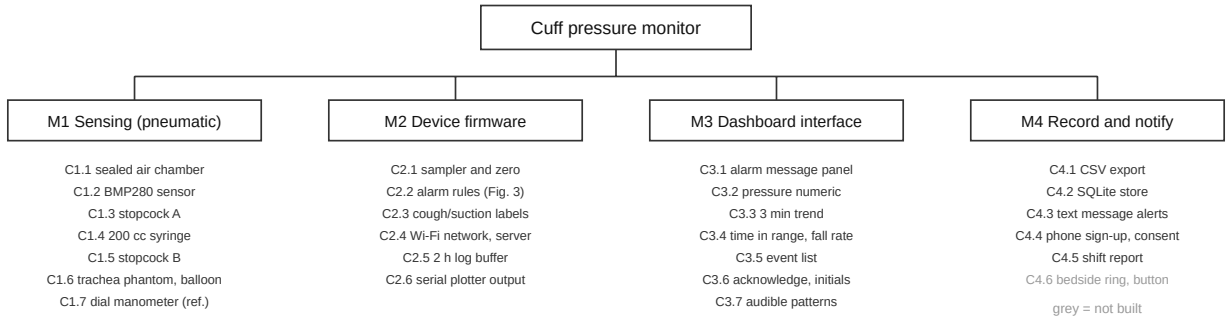


Figure 2. Component, module and system hierarchy for revision B. Components in grey were planned in the card and not built.

MODULE	STATUS	EVIDENCE
M1 Sensing	**BUILT, PARTIAL**	Two stopcock circuit built and used on 3 November. Manometer comparison observed, not recorded.
M2 Device firmware	**BUILT, PARTIAL**	C2.1, C2.2, C2.4 to C2.6 ran on hardware. C2.3 added on 5 November, tested in simulation only.
M3 Dashboard interface	**BUILT, PARTIAL**	Revision A used on hardware. Revision B (section 7) tested in simulation only.
M4 Record and notify	**BUILT, PARTIAL**	CSV export used on hardware. SQLite store, text alerts and shift report added on 5 November, tested in simulation with messages in dry run. Bedside indicators not built.

SECTION 5

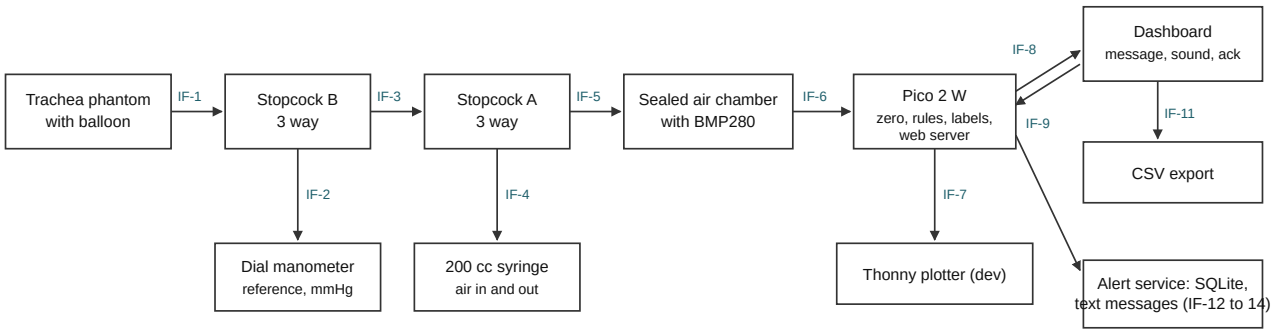
Interfaces

ID	BETWEEN	DIRECTION	SIGNAL, PROTOCOL AND RANGE	KIND
IF-1	Trachea balloon to stopcock B	both	Air through tubing, 0 to 60 cmH2O	Pneumatic
IF-2	Stopcock B to dial manometer	B to manometer	Air through tubing, read in mmHg, 20 to 300 mmHg dial	Pneumatic
IF-3	Stopcock B to stopcock A	both	Air through a direct Luer connection, 0 to 60 cmH2O	Pneumatic
IF-4	Stopcock A to 200 cc syringe	both	Air entry and exit, balloon inflation, 0 to 200 cc, 0 to 60 cmH2O	Pneumatic

IF-5	Stopcock A to sealed air chamber	both	Air, the sensed volume, 0 to 60 cmH2O in use; sensor limit about 88 cmH2O above room air	Pneumatic
IF-6	BMP280 to Pico	both (Pico is master)	I2C at 100 kHz, address 0x77, SDA on GP4, SCL on GP5, 3.3 V from 3V3 OUT through the breadboard rail, read 10 times per second	Digital
IF-7	Pico to Mac	Pico to Mac	USB serial, 10 lines per second for the Thonny plotter, development only	Serial
IF-8	Pico to browser	Pico to browser	Pico's own Wi-Fi network CuffMonitor, HTTP, JSON polled twice a second at 192.168.4.1	Network
IF-9	Browser to Pico	browser to Pico	Acknowledgement as HTTP POST with up to 12 letters of initials, one per press, shown within 0.5 s	Network
IF-10	Dashboard to computer speakers	browser to speakers	Browser audio, IEC 60601-1-8 style: high 880 Hz 10 beeps every 5 s, medium 660 Hz 3 beeps every 7 s, low 520 Hz 2 beeps	Audio
IF-12	Pico to laptop alert service	Pico to service	HTTP JSON over CuffMonitor, /data every 1 s, /contacts every 10 s (also the heartbeat)	Network
IF-13	Alert service to SQLite	service to file	cuff.db, one writer, readings at 1 Hz, every event, ack and message	File
IF-14	Alert service to Twilio	service to provider	HTTPS POST, up to 300 characters per text, 3 tries with backoff, status stored	Network
IF-11	Dashboard to CSV file	browser to file	Pressure, state, in range flag and events, one row per second, up to 2 h	File

SECTION 6, DELIVERABLE D4

System diagrams



Air path left of the Pico, electrical and network path right of it. Stopcock levers choose which branches share the closed volume.

Figure 3. Block diagram of the final bench configuration. Interface identifiers match section 5. The alert service runs on the laptop and was added on 5 November.

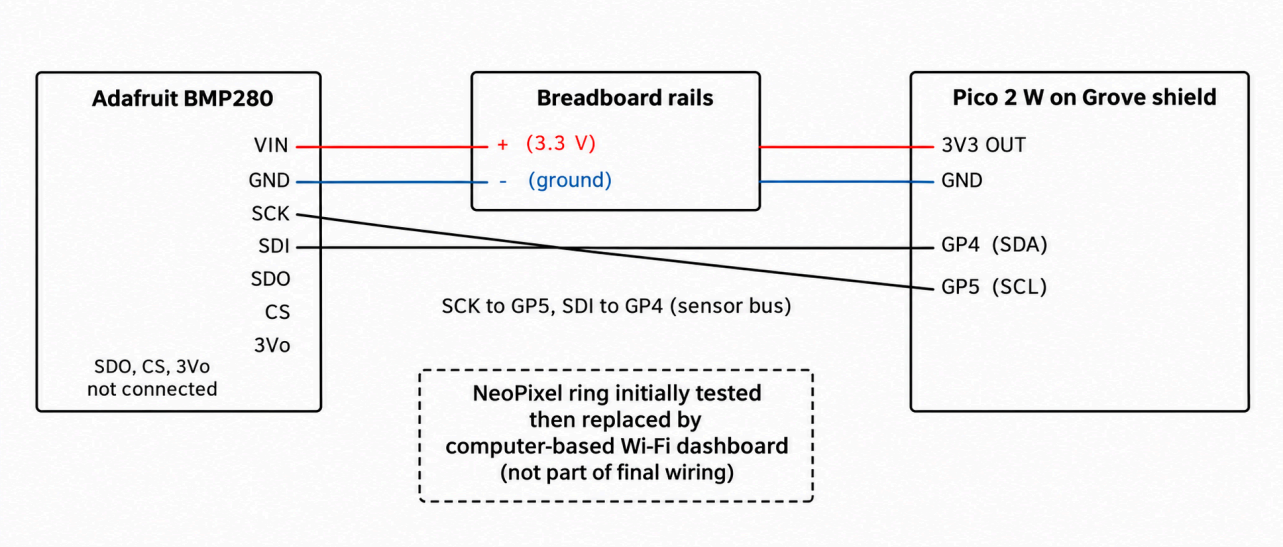


Figure 4. Wiring by pin name, as built and used in the bench run on 3 November 2026. VIN to 3V3 OUT and GND to GND through the breadboard rails, SCK to GP5, SDI to GP4.

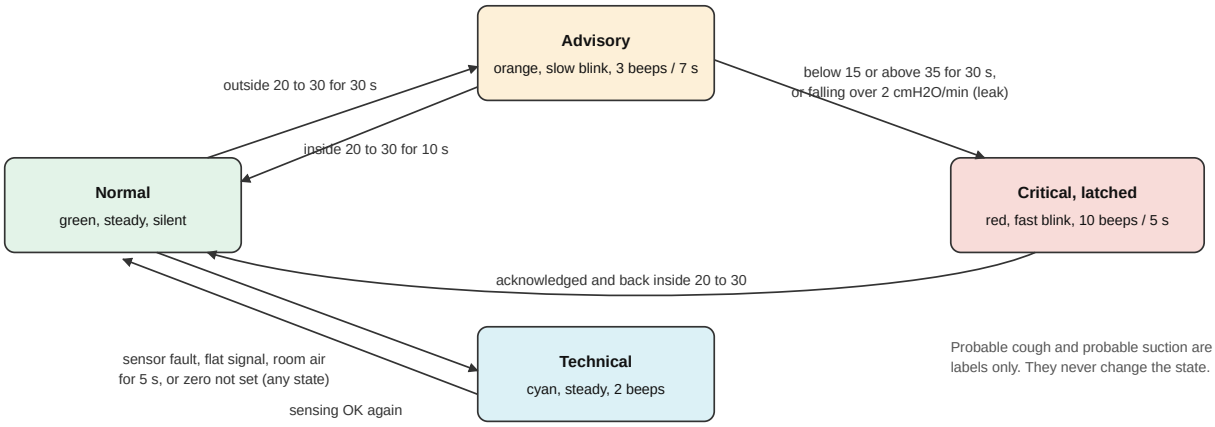


Figure 5. Alarm state chart as implemented in firmware v8, adapted from Figure 3 of the project card. The latch means a critical alarm clears only when it is acknowledged and the pressure is back in range.

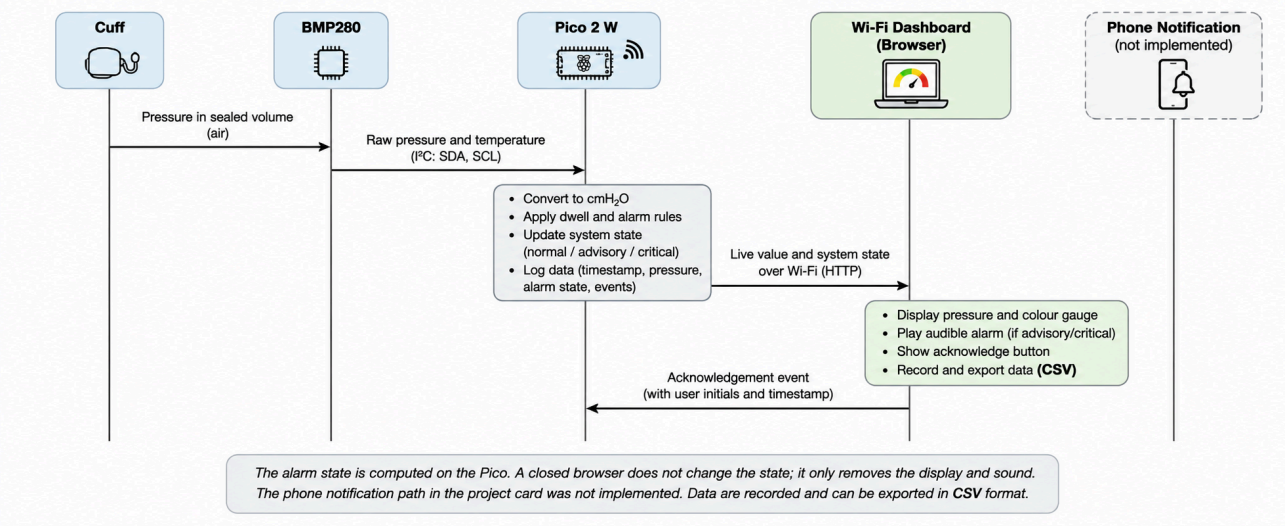


Figure 6. Sample to acknowledgement through the Wi-Fi dashboard. The alarm state is computed on the Pico, so a closed browser does not change it. This diagram shows revision A; revision B adds the alert service, which reads the same /data and texts each alarm (section 7, Figure 8).

SECTION 7, DELIVERABLE D5

Implementation

Firmware

The firmware is a single CircuitPython script, cuff_pressure.py, on CircuitPython 10.3.1 (the card lists version 9; 10.3.1 is what the board runs). Version v8 adds the phone sign-up routes for the alert service. It uses the adafruit_httpserver library and reads the BMP280 registers directly. `[zip: firmware/cuff_pressure.py]` It does the following:

- finds the BMP280 on GP4 and GP5 and waits, with a message on the dashboard, if it is not found;
- averages 20 readings over 2 s as the room air zero, so the syringe must be off at start-up;
- samples at 10 Hz, converts to cmH2O, and applies an optional calibration gain and offset;
- applies the Figure 5 rules: advisory after 30 s outside 20 to 30, critical after a further 30 s below 15 or above 35, leak when pressure falls faster than 2 cmH2O per minute over both the last 60 s and the last 20 s, technical state for a silent sensor, a flat signal, room air for 5 s or no zero;
- labels a rise of more than 5 cmH2O above the recent baseline that returns within 5 s as a probable cough, and a drop of more than 5 cmH2O that returns within 10 s as a probable suction pass. The labels are logged but never change the alarm state, so CR-2 still holds;
- creates the CuffMonitor network, serves the dashboard, and keeps a 2 h one row per second log for export.

```
class Transient:                    # cough and suction labels, firmware v8
    def update(self, now, p, ok, dt, alarm):
        d = p - self.base                          # distance from the slow baseline
        if not self.kind:
            if self.base >= 15 and d > COUGH_RISE:     self.kind = "COUGH"     # +5 cmH2O
            elif self.base >= 15 and d < -SUCTION_DROP: self.kind = "SUCTION"  # -5 cmH2O
            else: self.base += (p - self.base) * min(1.0, dt / BASELINE_TAU)     # 10 s
            return
        if abs(d) <= TRANSIENT_BACK:                # back within 2.5 cmH2O: event over
            alarm.log(now, "PROBABLE %s - no alarm" % self.kind)
            self.kind = ""
        elif now - self.start > (5 if self.kind == "COUGH" else 10):
            self.kind = ""                         # too long: the Figure 5 rules decide
```

Design modification: dashboard revision B

The first dashboard (revision A, used on 3 November) was a white web page with a colour circle, a banner and several cards that needed scrolling. It has been changed to look and behave like a bedside monitor and to fit one screen. It has a black background and a panel grid. One large message at the top gives the state in words: NORMAL in green, PROBABLE COUGH and PROBABLE SUCTION in orange, LOW or HIGH CUFF PRESSURE in orange, and AIR LEAK DETECTED or CRITICAL LOW or HIGH PRESSURE in red. Orange and red messages fill the panel with colour and blink, and the alarm tone sounds. Below the message are the pressure number, a 3 minute trend with the safe window shaded, time in range, the fall rate, recent events, and one row of controls: Acknowledge with initials, sound, test sound and CSV export.



Figure 7. Dashboard revision B in four states, from the desktop simulation of firmware v8 at 1280 by 800 pixels: normal, probable cough, probable suction, and air leak detected. Blinking is paused for the capture. Full resolution image: `[zip: docs/dashboard_states.png]`

1. Safety. The alarm rules stay on the Pico, so the dashboard only displays a state it is given and cannot invent or clear one. Red is kept for conditions that need action now (critical pressure and a leak) and blinks fast with the high priority 10 beep pattern until it is acknowledged. Orange is used for conditions to watch and blinks slowly with the medium priority pattern, which follows the priority coding of IEC 60601-1-8. Cough and suction are shown in orange with one short low priority tone but never raise an alarm, which keeps CR-2 and avoids alarm fatigue, the main reason staff silence monitors. A leak is named as a leak, not as low pressure, because the response differs (CR-3). After acknowledgement the panel stays red but stops blinking and goes silent, so a latched alarm is never hidden. If the browser loses the Pico, the page says NO SIGNAL FROM MONITOR and beeps, instead of freezing on the last good value (FM-7).

2. Usability. Everything fits on one laptop screen with no scrolling, so nothing important is below the fold. The layout matches the bedside monitors staff already use: a black background, labelled panels, a large coloured number with its high and low limits beside it, and a trend. Only one control is red and enabled at a time (Acknowledge, during a critical alarm). Initials are typed once and stored with the acknowledgement for CR-4. The orange sound button reminds the user that browsers block sound until the page is clicked once.

3. Clearness of the message. The state is written in plain words in the largest text on the screen, readable from across a room, so the nurse does not have to decode a colour or a number. Colour, blinking rate and sound all repeat the same priority, so the message still gets through to someone who is colour blind or not looking at the screen. A second line

says what to do or check, for example "Check pilot line, valve and cuff". The pressure number has its own colour scale, so a value outside 20 to 30 is visible even before the 30 s dwell raises an alarm.

Data recording and export

The Pico keeps the last 2 h of pressure and state, one row per second, plus up to 300 events, and the Export CSV button downloads them as a file that opens in Excel. `[zip: data/]` The two bench runs of 3 November produced 68 and 91 rows (section 9). The SQLite store and shift report below add durations of every excursion.

Database and text message alert path

A small alert service, cuff_service.py `[zip: service/cuff_service.py]`, runs on the laptop with Python's standard library only. Every second it reads the Pico's live values, stores them in a SQLite database, and stores each new event once. Every alert (advisory, critical, air leak, technical, probable cough or suction) is texted to every registered phone through Twilio, the messaging provider listed on the card. Each message is logged with its status, provider id and any error. If no Twilio keys are set, the service runs in dry run and logs the messages without sending them. Phones are registered on the dashboard itself: the SMS ALERTS button opens a form for a name, a mobile number and a consent tick box, and the consent time is stored, as the card's no identifiers rule requires. Messages carry no patient information. The command `python3 cuff_service.py report` prints the shift report for CR-4 from the database: time inside 20 to 30 cmH2O, every excursion with its duration and extremes, every alarm, and who acknowledged it.

```
readings(ts_utc, pico_t, cmh2o, ok, state, label)
alarm_events(event_id, ts_utc, pico_t, kind, text, cmh2o)
acks(event_id, ts_utc, user_initials, method)
users(phone_e164, name, optin_ts_utc, active)
notifications(id, event_id, phone_e164, ts_utc, body, status, provider_msg_id, error)
```

```
for (phone,) in db.execute("SELECT phone_e164 FROM users WHERE active=1"):
    status, sid, err = send_sms(phone, body)          # Twilio REST, 3 tries with backoff
    db.execute("INSERT INTO notifications(event_id, phone_e164, ts_utc, body, status,"
               " provider_msg_id, error) VALUES(?,?,?,?,?,?,?)", (eid, phone, utc(), body, status, sid,
err))
# body: "CUFF MONITOR (bench prototype): AIR LEAK DETECTED at 21:14:05, cuff 16.9 cmH2O. ..."
```

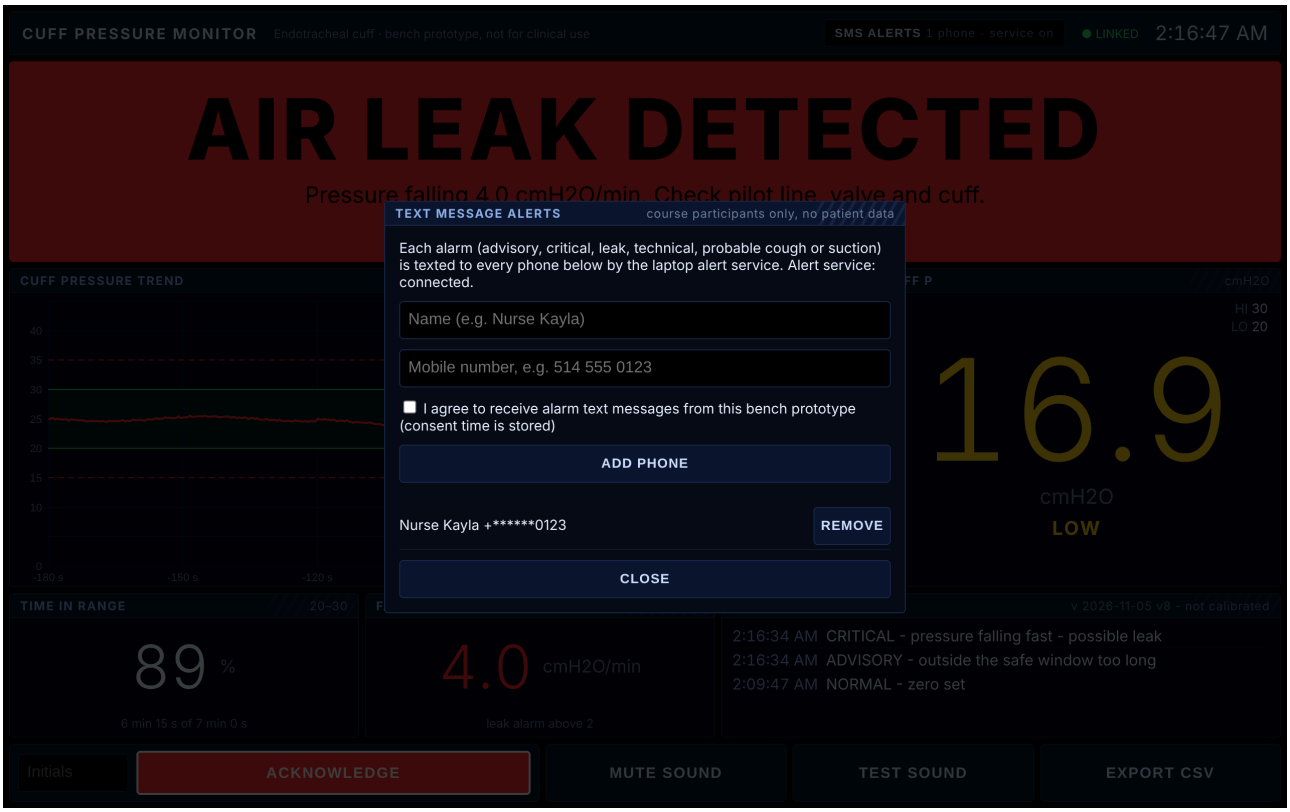
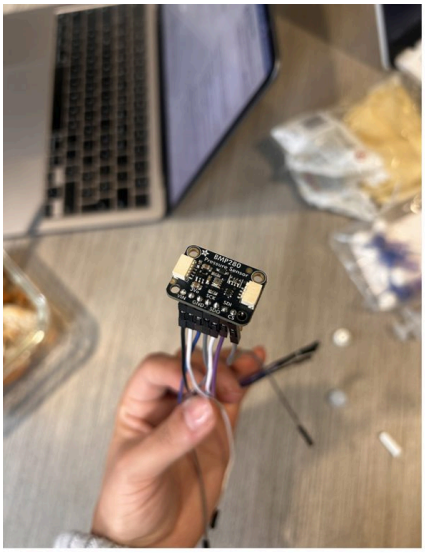


Figure 8. Text message sign-up on the dashboard (simulation). The top bar shows how many phones are registered and whether the alert service is connected, so a nurse can see when texts would not go out. Full resolution image: `[zip: docs/dashboard_sms_signup.png]`

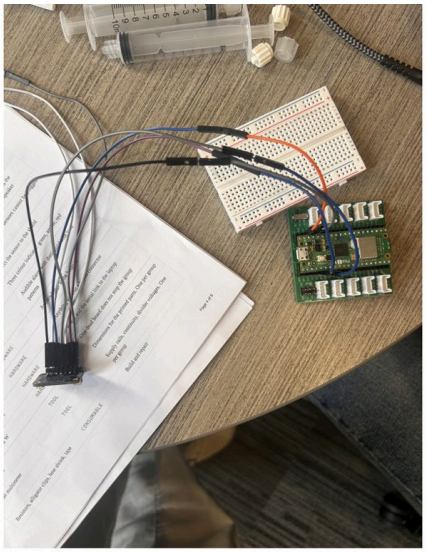
The laptop needs internet for the texts while it stays on CuffMonitor to reach the Pico, for example an iPhone on USB with Personal Hotspot. The local alarm on the dashboard does not depend on the service or on the internet.

Build record

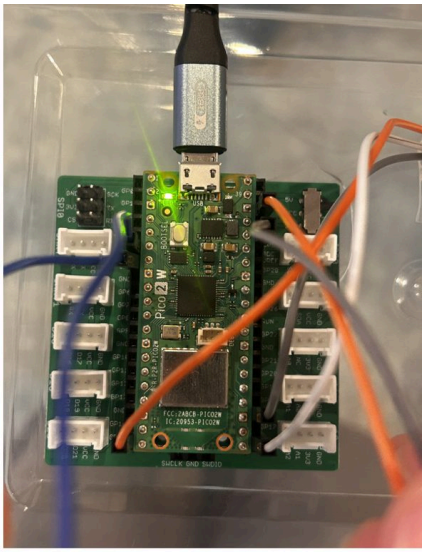
The photos below were taken during the build sessions and show the steps behind the design changes in section 3 and the failure modes in section 10.



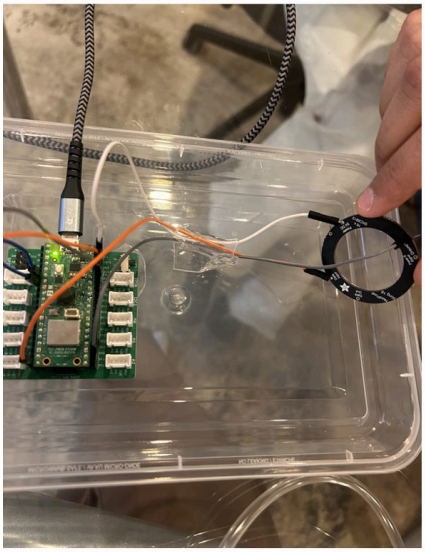
a BMP280 wired, 22 Oct



b Pico, breadboard, syringes, 22 Oct



c Pico 2 W on the Grove shield, 29 Oct



d NeoPixel ring test, 29 Oct



e Multimeter check (FM-3), 29 Oct



f Ring pins re-seated (FM-1), 29 Oct

Figure 9. Build record, 22 to 29 October 2026. (a) BMP280 with its four I2C wires, (b) first bench layout with the Pico, breadboard and syringes, (c) Pico 2 W on the Grove shield with the sensor wired to GP4 and GP5, (d) NeoPixel ring test, (e) multimeter check before the short circuit recorded as FM-3, (f) ring pins re-seated during the FM-1 investigation. The ring was later replaced by the dashboard.

Printed parts and repository

No 3D printed part is used in revision B. The repository holds cuff_pressure.py, README.md with wiring, set-up, run steps and troubleshooting, CHANGELOG.md, the alert service, the bench data file, and the desktop simulation used in section 9.3. It is published at github.com/imenbenadda/digitalprototypeproject_exsu620, and the same files are in the submission zip (Appendix A). Set-up and run steps are in the README `[zip: README.md]` and every change by date is in the changelog `[zip: CHANGELOG.md]`.

Deployment

1. Copy cuff_pressure.py to the Pico and run it from Thonny with stopcock A open to room air and the syringe off. Wait 2 s for the zero.
2. Wait about 10 s for the Shell to print that the CuffMonitor network is up.
3. Join CuffMonitor (password cuffpressure) from the laptop, open http://192.168.4.1, and click the page once to turn on sound.
4. The laptop has no internet while it is on CuffMonitor. This is expected.
5. Inflate the balloon with the 200 cc syringe through stopcock A, then turn the levers to close the syringe port.
6. Export the data with Export CSV at the end of the run.

SECTION 8, DELIVERABLE D6

Verification plan

ER	METHOD	APPARATUS	SAMPLE	ACCEPTANCE	SIGN-OFF
ER-1	Log inspection	BMP280, Pico 2 W, dashboard, CSV log	90 s recording	Rate within 10 percent of 1 Hz, step 0.1 cmH2O or less	Karman Kang
ER-2	Comparison against manometer	Dial manometer on stopcock B, 200 cc syringe, BMP280	At least 5 pressure levels, 3 repeats	No point over 2 cmH2O difference	Nathalie Obi
ER-3	Baseline stability	BMP280, Pico 2 W, line open to room air	At least 15 s after zero	Variation under 1 cmH2O	Imen Benadda
ER-4	Alarm dwell analysis	Syringe, BMP280, dashboard, CSV log	1 run with advisory and critical	Transitions within 2 s of the 30 s dwell	Kayla Arab
ER-5	Short excursion test	Syringe, BMP280, dashboard, CSV log	Rapid fluctuations in 1 run	No state change for events under 5 s	Lucy Han
ER-6	Staged leak	Syringe, slightly open stopcock port, dashboard, CSV log	1 run with a slow controlled fall	Labelled as a leak, not as low pressure	Karman Kang
ER-7	Technical alarm test	BMP280, Pico 2 W, dashboard, CSV log	1 run with a technical state	Technical within 10 s, no false normal	Nathalie Obi
ER-8	CSV reconstruction	Dashboard, CSV log	1 run (91 rows over 98 s)	Events and acknowledgements within 2 s	Imen Benadda

SECTION 9, DELIVERABLE D7

Verification report

9.1 What was observed on hardware

- The BMP280 was wired VIN to 3V3 OUT, GND to GND, SCK to GP5 and SDI to GP4, and live values were plotted in the Thonny plotter.
- On 3 November the alarm rules and the revision A dashboard were run on the bench. Technical, advisory and critical states were raised, the dashboard gave visual and audible alarms, and an acknowledgement was recorded.
- Two runs were exported as CSV with pressure, alarm state and event times: run 1 at 17:00:39 (68 rows over 72.5 s) and run 2 at 17:02:04 (91 rows over 97.8 s). In run 2 negative pressures were produced on purpose by pulling back the syringe.
- The NeoPixel ring did not work and was replaced by the dashboard.

9.2 Bench runs and comparison against the manometer

Pressure was varied by hand and the BMP280 readings were compared by eye with the dial manometer on stopcock B. The readings were observed to be consistent. The individual reference values were not recorded, so no error figure can be calculated and ER-2 is reported as not verified.

In run 1 the line stayed open to room air for 47 s after the zero and the reading stayed between -0.04 and +0.04 cmH2O. The balloon was then inflated to 22.6 cmH2O and the pressure fell to 1.1 cmH2O within 22 s, mostly in the first 5 s. This is the tubing leak that could not be sealed (FM-12) and that led to the two stopcock, 200 cc syringe design in section 3. The run ended before the 30 s advisory dwell, so no advisory or leak alarm was due.

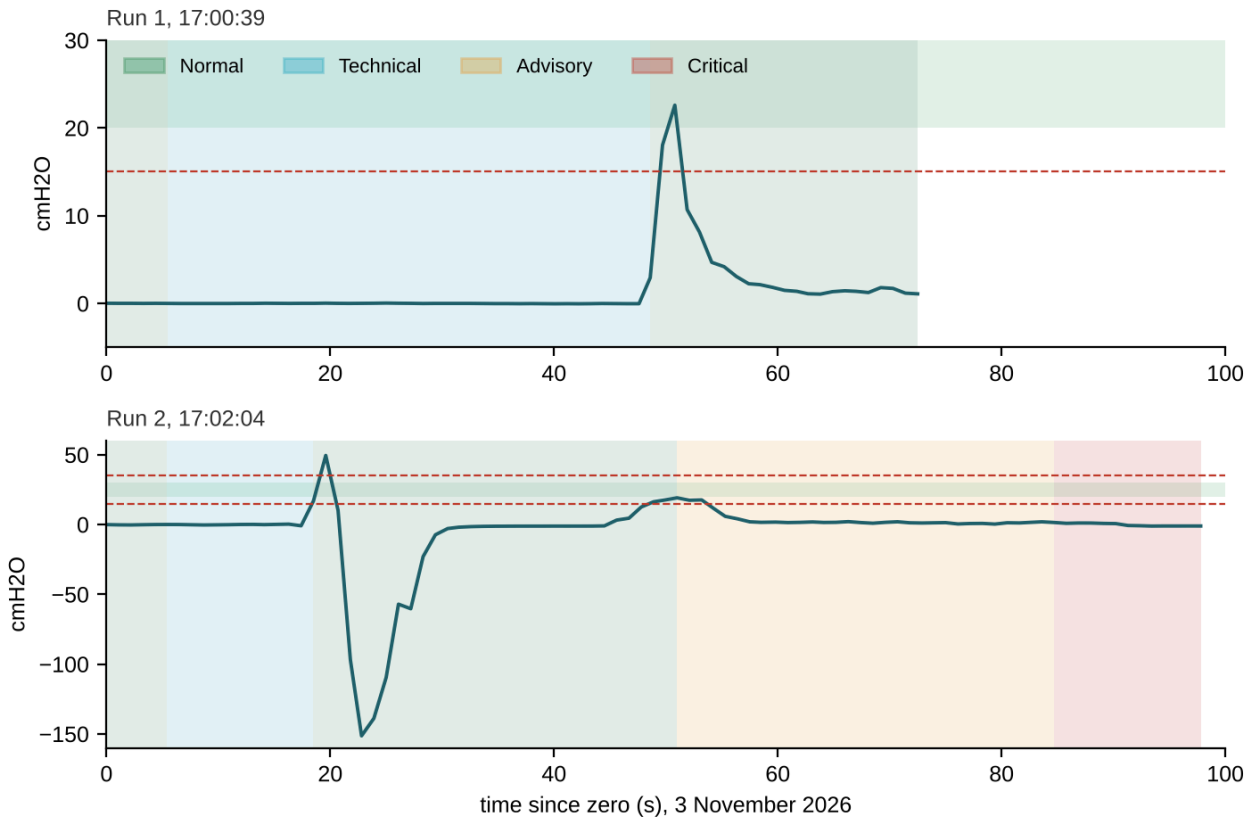


Figure 10. The two bench runs of 3 November 2026 from the exported CSVs `[zip: data/cuff_log_2026-11-03_1701.csv, data/cuff_log_2026-11-03_1703.csv]`, with the alarm state shaded and the safe window in green. Run 1 shows the leak after inflation; in run 2 the large negative swing is the syringe being pulled back. Note the different vertical scales.

MEASURE	WHAT THE LOG SHOWS	RELEVANT TO
Sample rate and precision	Mean interval 1.08 s in run 1 and 1.09 s in run 2 (0.92 Hz), within 10 percent of 1 Hz. Values logged to 0.01 cmH2O.	ER-1
Baseline variation	Run 1: -0.04 to +0.04 cmH2O for 47 s on room air. Run 2: -0.35 to +0.25 cmH2O for about 16 s.	ER-3
Technical state	Raised 4 s after the zero on room air in both runs (17:00:43 and 17:02:08). Cleared when pressure was applied (17:01:26 and 17:02:21). No false normal while raised.	ER-7
Tubing leak	Run 1: 22.6 cmH2O at 17:01:29, 4.7 cmH2O 4 s later, 1.1 cmH2O at 17:01:51.	FM-12
Rapid fluctuations	A peak of 49.5 cmH2O and a trough of -151.4 cmH2O during syringe handling (17:02:22 to 17:02:32), with no alarm.	ER-5
Advisory	Raised at 17:02:54, about 33 s after the technical state cleared, with pressure outside 20 to 30 cmH2O.	ER-4

Critical	Below 15 cmH2O from 17:02:58. Critical raised at 17:03:27, 29 s later, within one sample of the 30 s dwell.	ER-4
Acknowledgement	Acknowledged at 17:03:38. The alarm stayed red because pressure was still out of range.	ER-8
Leak label	Pressure fell from 17.6 to 1.5 cmH2O in about 5 s (17:02:57 to 17:03:02). Raised as a low pressure critical, not as a leak.	ER-6
Resting offset	After the negative swing the reading settled near -1.15 cmH2O rather than 0. Not yet known whether this is air left in the closed line or a sensor offset.	ER-2, FM-8

9.3 Desktop simulation of firmware v8

The cough and suction labels and dashboard revision B were added after the bench run, so they were tested by running the firmware's own Alarm and Transient classes on a computer with scripted pressure traces at 10 Hz, with 0.08 cmH2O of added noise. This checks the logic only, not the sensor or the Wi-Fi link. `[zip: tools/sim_cuff_pressure.py]` The test can be repeated with the command in README section 8.

SCRIPTED TRACE	RESULT	EXPECTED
25 cmH2O with a 2 s rise of 15 cmH2O	Labelled probable cough, state stays Normal	**AS EXPECTED**
25 cmH2O with an 8 s drop of 12 cmH2O	Labelled probable suction, state stays Normal	**AS EXPECTED**
Rise to 40 cmH2O and stay	Label dropped after 5 s, then Advisory and Critical at the 30 s dwell	**AS EXPECTED**
Slow fall at 4 cmH2O per minute from 25	Advisory, then Critical with the leak reason, shown as AIR LEAK DETECTED	**AS EXPECTED**
Step down to 10 cmH2O and stay	Critical low pressure, not a leak	**AS EXPECTED**
Alert service against the simulated Pico, one phone registered, scenarios cough, suction and leak	19 readings, 4 events and 3 texts stored in SQLite (dry run); shift report listed the excursion with its duration	**AS EXPECTED**
Critical, acknowledged, then back to 25	Returns to Normal; the recovery rise was briefly labelled a cough	**SEE FM-14**

9.4 Result against every requirement

ID	RESULT	VERDICT

ER-1	Two runs, 68 and 91 measurements, mean interval 1.08 and 1.09 s (0.92 Hz), within the 10 percent tolerance. Values logged in 0.01 cmH2O steps.	**PASS**
ER-2	Readings observed consistent with the dial manometer, but no paired values were recorded. The reference also cannot verify this requirement: the dial starts at 20 mmHg (27 cmH2O), so it cannot read 10 to 27 cmH2O, including the lower half of the safe window, and its 2 mmHg graduations (2.7 cmH2O) are coarser than the 2 cmH2O tolerance. A water column or digital cuff manometer, which was not provided in the kit (section 3), is needed. This is a limit of the equipment provided, not of the design.	**NOT VERIFIED**
ER-3	Variation 0.08 cmH2O over 47 s (run 1) and 0.6 cmH2O over 16 s (run 2) with the line open to room air, under the 1 cmH2O limit in both. Long term drift over a shift is not covered by this test and stays open as FM-11.	**PASS**
ER-4	Advisory and critical raised on hardware, critical 29 s after its condition. One run only.	**PARTIAL**
ER-5	Fast fluctuations raised no alarm on hardware. Durations not measured. Labels checked in simulation (9.3).	**PARTIAL**
ER-6	Not tested on hardware. The only fall in the bench run was a sudden drop from 17.6 to 1.5 cmH2O in 5 s, which is correctly a low pressure critical: the leak rule needs a sustained fall over 60 s. No slow controlled leak was staged. The rule raised AIR LEAK DETECTED in simulation (9.3).	**NOT VERIFIED**
ER-7	Technical state raised within 4 s and cleared on pressure in 2 of 2 runs. No deliberate fault injection (sensor unplugged, frozen signal).	**PARTIAL**
ER-8	91 rows and alarm events exported with an acknowledgement time on hardware. The SQLite store with excursion durations and acknowledging initials works in simulation only.	**PARTIAL**

SECTION 10, DELIVERABLE D8

Failure modes and mitigation

Severity (S), occurrence (O) and detectability (D) are rated 1 to 5, and the risk priority number (RPN) is their product. Ratings were agreed by the five team members; anything above 40 is either mitigated or declared open. FM-1 to FM-5 and FM-12 were found during building and troubleshooting.

ID	FAILURE MODE	EFFECT	S	O	D	RPN	MITIGATION	STATUS
FM-1	Ring ground wire resting beside the pad, not in it	No bedside visual alarm	4	3	3	36	Ring replaced by the dashboard	Closed by design change
FM-2	Ring power and ground on wrong pins	Ring or Pico damaged	4	2	2	16	Wire by pin name from a checklist, power off before rewiring	Pico recovered, ring no longer used

FM-3	Short circuit during multimeter testing	Pico off, monitoring stops	5	2	3	30	Meter set-up check before probing; dashboard shows NO SIGNAL when the Pico stops	Pico recovered; NO SIGNAL tested in simulation
FM-4	Thonny ran an older script than the one documented	Device differs from the dossier	3	3	4	36	Version printed at start and shown on the dashboard; CHANGELOG and _originals kept	Implemented
FM-5	Library missing on the board	Firmware fails to start	3	2	1	6	Library list and versions in the README	Implemented
FM-6	Browser blocks sound until the page is clicked	Silent alarm	5	4	3	60	Orange sound button until clicked, Test sound button	Implemented; a click is still required
FM-7	Laptop closed or leaves CuffMonitor	No alarm reaches the nurse	5	3	3	45	Rules on the Pico; page shows NO SIGNAL and beeps when the link drops	OPEN: no independent bedside alarm
FM-8	Zero taken with the syringe attached	Every reading offset	4	3	4	48	Zero only with stopcock A open to room air	Procedure used; not fault tested
FM-9	Cough or suction raises an alarm	Alarm fatigue	3	4	2	24	Dwell times; cough and suction shown as labels only	Hardware: no alarm. Labels: simulation
FM-10	Slow leak read only as low pressure	Wrong clinical response	4	3	3	36	Separate fall rate rule with its own message, AIR LEAK DETECTED	Simulation only; not verified on hardware
FM-11	Room pressure or temperature changes after the zero	Reading drifts with no cuff change	4	4	5	80	Second BMP280 open to room air, or scheduled re-zero	OPEN: not implemented
FM-12	Air leaks in the tubing and connectors	Pressure cannot be held, readings fall	4	5	3	60	Two stopcocks and a 200 cc syringe to draw air back and re-inflate	Leak measured in run 1 (22.6 to 1.1 cmH2O in 22 s); design changed; leak not removed
FM-13	Manometer left connected during a logging run	Added volume changes the pressure measured	3	4	3	36	Turn stopcock B away from the manometer during logging runs	OPEN: procedure to adopt
FM-15	Text message not delivered (no internet, provider error, trial limits)	Nurse away from the screen not alerted	4	3	3	36	Retry 3 times, status and error stored per message, service status shown on the dashboard, local alarm unaffected	Dry run only; real sending not tested
FM-14	Re-inflation by the nurse read as a cough	Brief wrong label	2	3	2	12	Labels only above 15 cmH2O baseline; label never alarms	Seen in simulation, accepted

SECTION 11, DELIVERABLE D9

Traceability and limitations

CLINICAL REQUIREMENT	ENGINEERING REQUIREMENTS	EVIDENCE	STATUS
CR-1	ER-1, ER-2, ER-3	Section 9.4: ER-1 and ER-3 pass on hardware; ER-2 not verified because manometer values were not recorded	PARTIAL
CR-2	ER-4, ER-5	Sections 9.2 to 9.4: advisory and critical with 30 s dwell, no alarm on fast fluctuations	PARTIAL
CR-3	ER-6, ER-7	Sections 9.2 to 9.4: technical alarm and recovery on hardware; leak label in simulation only	PARTIAL
CR-4	ER-8	Sections 7 and 9: CSV export with an acknowledgement time on hardware; SQLite shift report with durations and initials in simulation	PARTIAL

Declared limitations

- Equipment: the kit contained only a balloon, a trachea phantom and a manual blood pressure cuff with a dial manometer. Tubing, syringes and stopcocks were bought by the team. The dial has no electronic output, reads mmHg from 20 mmHg (27 cmH2O) and cannot check the lower half of the safe window, so ER-2 could not be verified with the reference provided. This was outside the team's control (section 3).
- In the final configuration the manometer can stay connected during a run, which the card's measurement integrity rule forbids. Stopcock B must be turned away from it during logging runs (FM-13).
- The tubing still leaks. The 200 cc syringe works around the leak; it does not remove it (FM-12).
- The zero is taken once against room air at start-up. Room pressure changes over a shift may move the reading (FM-11).
- Only two short bench runs were recorded. The cough and suction labels, the leak message and dashboard revision B were tested in simulation, not on the board.
- The alarms depend on the laptop staying on the CuffMonitor network. There is no independent bedside alarm (FM-7).
- The SQLite store, shift report and text messages were tested in simulation, with messages in dry run. Real texts through Twilio have not been sent.
- The device is a bench prototype and has never been connected to a patient or to clinical equipment in service.

Technical memorandum, summary

The prototype measures the pressure inside a breathing tube cuff continuously, using a small barometric sensor in a sealed air chamber connected to the cuff through two valves. It subtracts the room pressure measured at start-up and reports the result in cmH2O, the unit of the cuff manometer. A small controller checks whether the pressure stays inside the recommended 20 to 30 cmH2O, has been outside for too long, is falling in a way that suggests a leak, or whether the sensor itself has a problem. A laptop or tablet on the controller's own Wi-Fi shows this on a black monitor style

screen, with one large message in plain words, colours and sounds that follow hospital alarm conventions, and a button to acknowledge alarms with initials. Short spikes and dips from coughing or suctioning are named on the screen but do not raise an alarm. In two bench runs the device measured continuously, raised its alarms at the right times and exported its data. The accuracy against the manometer, the leak alarm on real hardware and full shift reporting are not yet shown. It is a bench prototype, not a medical device.

Use of generative AI

In line with section 6.4 of the course handbook, the team declares that a generative AI assistant (Claude, Anthropic) was used for: writing and debugging the CircuitPython firmware run from Thonny on the Pico 2 W; building the Wi-Fi dashboard, including revision B; and the layout and editing of this dossier, including drafting revised wording for sections the team then reviewed. The clinical requirements, the physical build, the bench testing, the data, the design decisions and the judgements in this dossier are the team's own. The team checked the firmware on the bench and reviewed every statement, number and claim in this document, and takes responsibility for its accuracy.

Appendix A. Submission contents

Everything below is in EXSU620_Group2_Submission.zip, beside this PDF. The same files are in the public repository github.com/imenbenadda/digitalprototypeproject_exsu620.

FILE IN THE ZIP	WHAT IT IS	CITED IN
EXSU620_Group2_Design_Dossier.pdf	This design dossier, D1 to D9	-
README.md	Hardware, wiring by pin name, set-up, run steps, alert service, calibration steps, simulation, troubleshooting	Section 7
CHANGELOG.md	Every change to firmware, service and dossier, by date	Section 7
firmware/cuff_pressure.py	Firmware v8, the only file on the Pico: sensor, zero, alarm rules, cough and suction labels, Wi-Fi dashboard, SMS sign-up, 2 h log	Section 7
service/cuff_service.py	Laptop alert service: SQLite database, a text for every alert, shift report	Section 7
data/cuff_log_2026-11-03_1701.csv	Bench run 1, 68 rows: room air baseline, technical alarm, inflation and tubing leak	Section 9.2, Figure 10
data/cuff_log_2026-11-03_1703.csv	Bench run 2, 91 rows: technical, advisory, critical and acknowledgement	Section 9.2, Figure 10
tools/sim_cuff_pressure.py	Desktop simulation and the six logic tests of section 9.3; also serves the dashboard for testing	Section 9.3
docs/final_rig.jpg	Photos of the final two stopcock bench rig	Figure 1
docs/dashboard_states.png	Dashboard revision B: normal, probable cough, probable suction, air leak detected	Figure 7
docs/dashboard_sms_signup.png	Dashboard SMS ALERTS sign-up	Figure 8